

A Deterministic Weight-of-Evidence Framework for Firm-Level Shadow Economy Risk, with Confidence Decoupled from Direction

Shadow Index Project

Working Paper v3 · Prototype: synthetic demonstration data

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Abstract

We describe a two-part system for flagging firm-level involvement in Uzbekistan’s shadow economy from a prototype dataset: a deterministic weight-of-evidence scorecard that converts seven firm-level forensic contradiction checks into a probability, a risk score, and—critically—a confidence figure computed independently of the probability’s direction; and a downstream large-language-model layer that turns the scorecard’s output into a plain-language case narrative without ever touching the arithmetic. The central methodological claim is narrow and falsifiable: probability answers *which way* the evidence points, confidence answers *how much* and *how decisive* the evidence is, and conflating the two—which the system’s first implementation did—produces a number that looks precise but silently discards half of what the evidence actually says. We give the exact formulas, the calibration choices behind them, three fully traced worked examples against real code paths, and an explicit accounting of what in this pipeline is real mathematics versus illustrative demonstration data.

Keywords: shadow economy; weight-of-evidence scorecard; tax compliance risk; forensic accounting signals; uncertainty quantification; large language models.

1 The Problem with One Number

Every firm-level risk system eventually has to answer two questions that are easy to mistake for one: how likely is this firm to be involved in the shadow economy, and how sure are we? The first implementation of this system’s “AI Shadow Economy Analysis” card answered both with the same number, displayed twice—a 1–10 score derived from a probability, and that same probability shown again underneath the word *confidence*.

The failure mode is concrete. A clean firm with seven forensic checks all reading comfortably compliant would report a low probability (correctly) and, under the old design, a low “confidence” alongside it—as if having strong, complete evidence that a firm is *not* involved somehow made the system less sure. Conversely, a firm with two signals that had barely tipped over their statutory threshold would report the same inflated confidence as a firm where every check was decisively, unambiguously flagged. Confidence, in other words, was tracking the score’s magnitude, not the

evidence’s quality.

Section 5 gives the fix: confidence is computed from evidence *coverage* and *decisiveness*—quantities that do not know or care which direction the probability points. Sections 3–4 first establish what the evidence actually is and how it becomes a probability at all.

2 System Architecture

The pipeline has five parts. Four are deterministic arithmetic; the fifth is a schema-constrained language model that is structurally prevented from touching a single number the other four produce. Figure 1 gives the layout.

Layers B and C are not sequential stages—they are two outputs of one pass over Layer A’s signals (Sections 4–5). Layer D reaches the narrative layer through a deliberately secondary path: the system prompt (Section 7) explicitly forbids treating regional macro context as evidence against a specific firm.

This specification differs from an earlier draft for this project in one deliberate way. That draft named a positive-unlabeled (PU) learning classifier, Manski partial-identification bounds, and a Fay–Herriot small-area estimator as parts of the pipeline. None of those are implemented. This paper documents what runs today and files the remainder under Roadmap (Section 9) rather than presenting aspiration as architecture.

3 Layer A — Forensic Contradiction Signals

Every signal in Layer A tests a firm’s own filings against each other, or against a second, independently sourced record of the same event—never against a peer average. A firm’s declared revenue either matches its point-of-sale settlement volume or it does not; that is a same-firm arithmetic fact, not a statistical inference. This is what makes the signals usable as hard evidence rather than circumstantial pattern-matching. The mechanism behind the invoice-chain check in particular is the third-party paper trail documented by Pomeranz [6]; the wage-bunching check follows the threshold-bunching logic of Kleven and Waseem [3].

Table 1 lists the seven checks as implemented in `forensics.js`.

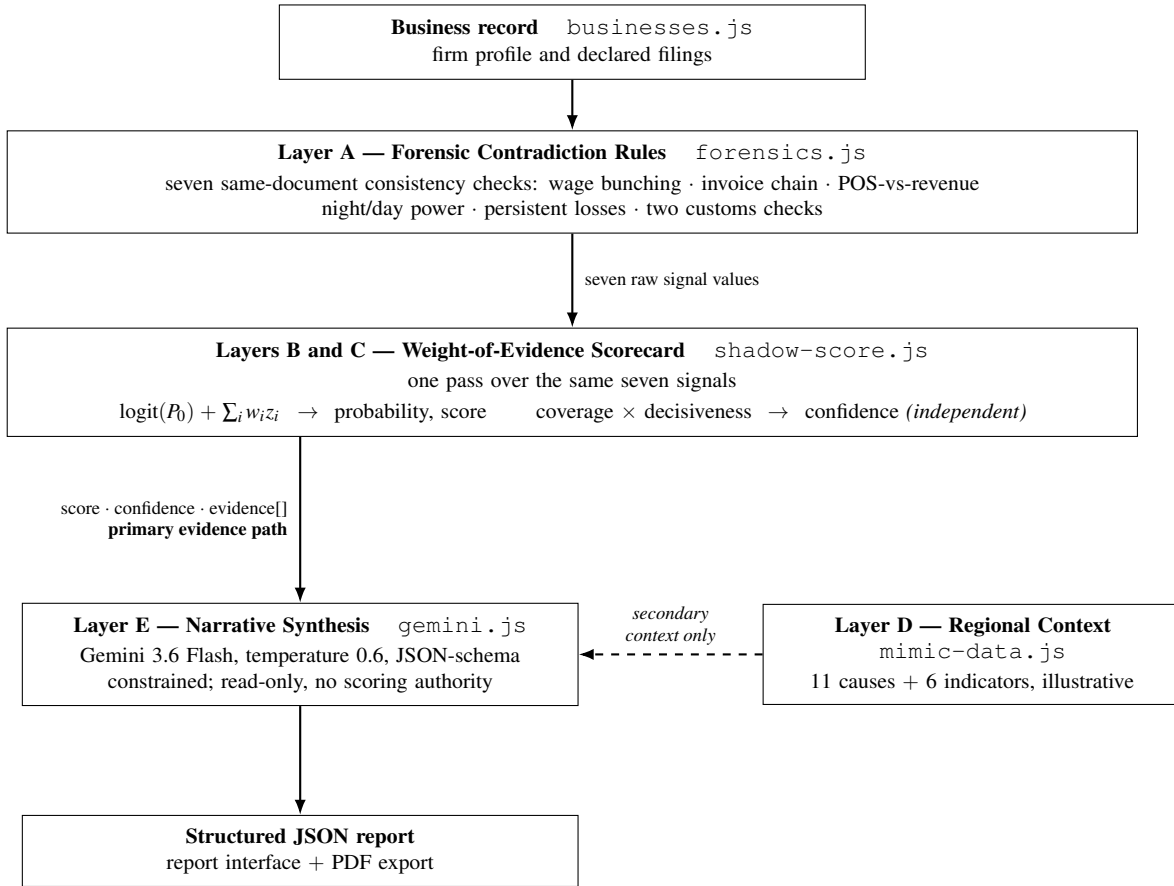


Figure 1: Pipeline architecture. Layers B and C are two outputs of a single pass over Layer A’s signals, not sequential stages. Layer D (dashed) reaches the narrative layer through a deliberately secondary path; the system prompt forbids treating regional macro context as evidence against a specific firm.

“Sector-gated” means the two customs checks apply only to the nine trade-exposed sectors (wholesale and retail, logistics, manufacturing, construction, textiles, mining, food processing, agriculture, automotive trade). A professional-services or software firm has no import declarations to check against, and the scorecard treats that as *missing* evidence rather than as a clean result. This distinction turns out to matter a great deal for confidence (Section 5).

Each signal’s flag boolean is the single source of truth for “contradiction found” throughout the system. The user interface’s forensic-signals card and the scorecard’s probability computation both read the same condition, so the two can never disagree about whether a line was crossed.

4 Layer B — The Scorecard

The scoring method is a weight-of-evidence logistic scorecard—the same family of model used in credit-risk underwriting [8] and recommended by the OECD for tax administrations before enough audit-verified outcomes exist to fit a full regression [5]. Each applicable signal contributes a log-odds term sized by how far past its statutory threshold the firm sits; the terms sum under a conditional-

independence assumption and pass through a logistic link.

4.1 Peer-cohort prior

Scoring starts from a prior tied to the firm’s baseline sector/region risk tier, not a single flat rate for every firm (Table 2). This is the direct fix for a detection-density bias—an unadjusted administrative index tends to flag whichever districts have the most digital paper trail to check, not the most actual informality—because it means a Critical-tier firm with clean signals settles back near its peers instead of being auto-flagged by tier alone.

4.2 Threshold, scale, and evidentiary weight

Each signal has a *threshold* (exactly the flag line of Table 1), a *scale* (how far past that line counts as one full unit of evidence), and a *weight* (evidentiary strength—higher for same-document contradictions, lower for peer-relative patterns that have innocent explanations on their own). Table 3 gives the values.

Table 1: The seven forensic contradiction signals (`forensics.js`). Each tests a firm’s own filings against each other or against an independently sourced record of the same event.

Signal	Tests	Flag condition	Sector-gated
Wage bunching	share of staff paid exactly minimum wage (1,050,000 UZS/mo)	share > 30%	no
VAT invoice chain	declared sale vs. counterparty’s declared purchase, same transaction	gap > 15%	no
Night/day electricity	off-hours vs. daytime industrial power draw ratio	ratio > 0.55	no
POS vs. declared revenue	card and fiscal-receipt volume vs. declared revenue	ratio > 1.4×	no
Persistent losses	consecutive loss-making years while still operating	≥ 3 yrs, status Active	no
Mirror trade statistics	customs-declared import value vs. UN Comtrade partner-country export	gap > 20%	yes
Customs value vs. COGS	customs-declared import value vs. declared cost of goods sold	gap > 20%	yes

Table 2: Prior probability by baseline risk tier.

Tier	Prior P_0	logit(P_0)
Low	4%	−3.178
Moderate	10%	−2.197
Elevated	20%	−1.386
Critical	33%	−0.708

Table 3: Scorecard parameters. Weight encodes evidentiary strength: higher for same-document contradictions, lower for patterns with plausible innocent explanations.

Signal	Threshold	Scale	Weight
POS vs. revenue	1.40	0.40	1.1
VAT invoice chain	0.15	0.15	0.9
Customs vs. COGS	0.20	0.20	0.8
Mirror trade	0.20	0.20	0.7
Wage bunching	0.30	0.20	0.5
Night/day electricity	0.55	0.20	0.5
Persistent losses	3 yrs	2 yrs	0.4

4.3 The formula

For each applicable signal i with raw value r_i , threshold t_i , scale s_i , and weight w_i :

$$z_i = \text{clip}\left(\frac{\max(0, r_i - t_i)}{s_i}, 0, 3\right) \quad (1)$$

$$c_i = w_i z_i \quad (2)$$

$$\ell = \text{logit}(P_0) + \sum_i c_i \quad (3)$$

$$P = \text{clip}(\sigma(\ell), 0.01, 0.99) \quad (4)$$

$$S = \text{clip}(\text{round}(1 + 9P), 1, 10) \quad (5)$$

where P_0 is the tier prior of Table 2 and $\sigma(\cdot)$ is the logistic function.

A signal at or below its threshold contributes exactly zero. “No evidence of a problem” is not the same as “evidence there is not one,” and the formula does not let a clean reading manufacture negative suspicion. The cap at $z = 3$ is a conservatism device: no single signal, however extreme, can alone drive the probability to certainty. Section 4.4

Table 4: Full contribution trace, Termiz Chegara Logistika (Critical baseline, all seven checks flagged). Live output of `computeShadowProbability()`.

Signal	raw	excess	z	wz
POS vs. revenue	3.31×	1.91	3.00 [†]	3.300
VAT invoice chain	55%	0.40	2.67	2.400
Customs vs. COGS	49%	0.29	1.45	1.160
Wage bunching	75.8%	0.46	2.29	1.145
Night/day electricity	91%	0.36	1.80	0.900
Mirror trade	45.1%	0.25	1.26	0.878
Persistent losses	6 yrs	3	1.50	0.600
Total				10.383

[†]Raw z would be 4.78; the cap at $z = 3$ engages here (Section 4.3).

shows it engaging.

4.4 Worked example — full derivation

Termiz Chegara Logistika (Critical baseline; Logistics and Transport; status Active) trips all seven checks. Every term in Table 4 is the live output of `computeShadowProbability()`.

Summing the contributions:

$$\begin{aligned} \ell &= \text{logit}(0.33) + 10.383 \\ &= -0.708 + 10.383 = 9.675 \\ P &= \sigma(9.675) \rightarrow \text{clipped} = 0.99 \\ S &= \text{round}(1 + 9 \times 0.99) = 10/10 \text{ (Critical)} \end{aligned}$$

5 Layer C — Confidence, Decoupled from Direction

Confidence answers a different question than probability: not *which way* the evidence points, but how much evidence exists and how decisively it sits away from the line separating “flagged” from “clean.” It is built from two quantities, computed over the same signal loop as Section 4 but summed independently of the sign of z .

5.1 Coverage

Coverage is the fraction of the seven possible checks that were even applicable to this firm. A non-trade-exposed

firm has no customs record to check and caps out at 5/7; a firm under review has its persistent-losses check gated off entirely (Section 3), capping it at 6/7. Fewer applicable checks means a narrower evidence base, independent of what the available checks say.

5.2 Decisiveness

For every applicable signal, decisiveness is an *unsigned* distance from the threshold. A value sitting exactly on the line is the weakest possible evidence regardless of which side it lands on; a value far from the line in either direction is strong evidence either way. The distance saturates through $1 - e^{-x}$ so that a couple of extreme signals cannot alone manufacture near-certainty:

$$d_i = \frac{|r_i - t_i|}{s_i}, \quad \delta_i = 1 - e^{-d_i} \quad (6)$$

$$\kappa = \frac{n_{\text{applicable}}}{7}, \quad \bar{\delta} = \frac{1}{n} \sum_i \delta_i \quad (7)$$

$$C = \text{clip}(\text{round}(100(0.3\kappa + 0.7\bar{\delta})), 10, 96) \quad (8)$$

The 70/30 split weights how-decisive over how-much-evidence: a firm with complete coverage but every value sitting exactly on its threshold line should not outrank a firm with slightly fewer applicable checks whose values are unambiguous. Confidence is clipped to [10, 96]—never absolute certainty, and never treated as fully uninformative even from a single decisive signal.

5.3 Three cases, traced

The point of this section is empirical, not merely definitional. Table 5 shows three businesses run through the live scorecard.

Read Toshkent and Termiz side by side: opposite ends of the probability scale, both landing at high confidence—because in both cases the evidence is complete and unambiguous, just pointing in different directions. Andijon is the case that actually stress-tests the design. A high score built partly out of two barely-tipped-over signals lands at meaningfully lower confidence than either of the decisive cases, despite outscoring Toshkent by seven points. A system that only ever showed probability would have no way to express that.

5.4 Why not simply widen a confidence interval?

A Wilson or Wald interval around the probability would conflate the same two things this section separates. Interval width in a logistic model is driven mechanically by how far the log-odds sits from zero, which is exactly the “which way” quantity, not an independent “how much evidence” quantity. Coverage and decisiveness were chosen because they are computable directly from what is actually missing or ambiguous in the evidence, rather than derived algebraically from the probability itself.

6 Layer D — Regional Macro Context

Independently of firm-level scoring, each of Uzbekistan’s 14 regions carries a MIMIC-inspired macro dataset: 11 “causes” (tax burden, labor-market rigidity, corruption control, unemployment, GDP per capita, and others) and 6 “indicators” (currency-demand ratio, large-denomination banknote share, electricity-consumption index, labor-force participation, real GDP growth) in the tradition of Schneider and Enste [7].

Every variable’s value is generated deterministically: linear interpolation between a literature-informed plausible range, positioned by the region’s existing shadow-index score, plus a small deterministic jitter keyed to the region and variable name so that figures are not perfectly collinear. This is worth stating with complete precision, because it is easy to imply more than it is. *No structural equation model is fitted*. There are no estimated factor loadings, no latent-variable regression, and nothing resembling the causal identification a real MIMIC study requires. The layer exists to give the narrative model (Section 7) regional texture to write about. It is a presentation dataset, not a statistical estimate, and it never contributes a term to the firm-level probability of Section 4.

Known limitation. Breusch [1] documents precisely how MIMIC-style claims mislead when the “model” is asserted rather than fitted against real covariance structure. Layer D avoids that failure mode by not claiming to be a fitted model in the first place—but a reader skimming the interface without this paper could still mistake “MIMIC causes/indicators” for real regional statistics. The system prompt in Section 7 is the only enforcement mechanism preventing that conflation from reaching the narrative output.

7 Layer E — Narrative Synthesis

A single Gemini 3.6 Flash call, constrained to a nine-field JSON schema (notes, executiveSummary, forensicFindings, causesAnalysis, indicatorsAnalysis, conclusion, estimatedIndex, confidencePercent, keyRiskFactors), turns the scorecard’s already-computed numbers plus Layer D’s regional context into a case narrative. It has no scoring authority: every number in the preliminary screening block handed to the model—score, probability, confidence, per-signal evidence—was computed in Sections 4–5 before the model ever runs.

The system instruction enforces the asymmetry Figure 1 draws visually, in three explicit rules:

1. Forensic signals (Section 3) are direct evidence about this specific firm; a single flagged signal should anchor the assessment.

Table 5: Three cases through the live scorecard. Toshkent and Termiz sit at opposite ends of the probability scale yet both report high confidence, because in both the evidence is complete and unambiguous. Andijon scores seven points above Toshkent yet reports lower confidence, because two of its flags sit within one point of their thresholds.

Firm	Evidence state	Score	P	Conf.	Coverage	Note
Toshkent Metall Profil	all 7 checks clean	2/10	10%	59%	86% (6/7)	losses gated (Under Review); avg. decisiveness 48%
Andijon Mashinasozlik Zavodi	flagged, but marginally	9/10	84%	54%	—	wage bunching 30.9% vs. 30%; COGS gap 20.3% vs. 20%
Termiz Chegara Logistika	all 7 decisively flagged	10/10	99%	89%	100% (7/7)	avg. decisiveness 84%

- MIMIC causes and indicators (Section 6) are regional context only; the model is instructed never to treat a macro variable alone as evidence against a specific firm.
- `estimatedIndex` must stay within 2 points of the scorecard’s own score unless a forensic signal is flagged, in which case the model may diverge further but must cite the specific signal driving the divergence.

Every call is logged—console line, local `.jsonl` file, and an in-memory ring buffer surfaced at `GET /api/usage`—with prompt, thought, and completion token counts plus latency, so cost and behavior are auditable in aggregate rather than only per report.

Known limitation. Temperature is set to 0.6, not 0, meaning the narrative can phrase the same underlying numbers differently across two runs on the same firm. An earlier draft specification called for temperature 0 plus a hard post-hoc check that every numeric token in the output exists in the input payload (“numeric token invariance”). Neither is implemented yet; it is the most concrete and cheapest item in the roadmap (Section 9).

8 Validation Status

What follows is an unusually direct accounting, by design: a scorecard whose limitations are asserted by the same document that built it is either honest or has not looked hard enough. Table 6 is the look.

The single largest gap is the one every weight-of-evidence scorecard starts with: no audit-verified labels exist yet against which to check the thresholds and weights of Table 3. The positive-unlabeled learning correction of Elkan and Noto [2]—the standard way to recover true posterior probabilities once some positives are confirmed and the rest are merely unlabeled rather than known-negative—is the natural next step the moment labeled cases exist, and would replace Table 3’s hand-set weights with fitted ones.

9 Roadmap

- Calibrate Table 3 on real outcomes.** Once audit-confirmed cases accumulate, fit weights via PU-learning [2] instead of literature-informed defaults.
- Numeric-token invariance on Layer E output.** Reject and regenerate any narrative containing a figure not

traceable to the input payload—cheap, mechanical, and closing the one place this pipeline currently tolerates non-determinism.

- Region-level partial identification.** Manski [4] bounds for aggregating firm-level lower bounds into a district total, rather than treating Layer D as free-standing context.
- Replace Layer D’s interpolation** with a fitted structural model, or relabel it more conservatively than “MIMIC” until one exists.
- Real administrative data feeds**—EHF e-invoicing, Humo/Uzcard POS settlement, ASKUE electricity telemetry, State Customs declarations—behind the statutory data-sharing agreements each requires, replacing the synthetic values in Table 6’s right-hand column one signal at a time.

References

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Table 6: What is real versus illustrative, by layer.

Layer	Mathematics	Underlying data
A — forensic signals	<i>real logic</i> : exact threshold comparisons	<i>synthetic</i> : deterministic per-business mock values
B — probability, score	<i>real</i> : logistic scorecard, reproducible	weights/thresholds are literature-informed defaults, not fitted to audit outcomes
C — confidence	<i>real</i> : coverage $\times$ decisiveness	0.3/0.7 split is a reasoned default, not empirically tuned
D — regional context	<i>illustrative</i> : interpolation, not a fitted SEM	<i>synthetic</i> : generated, not sourced from GRP/LFS statistics
E — narrative	schema-constrained generation	reasons over B–D’s outputs; temperature 0.6, no invariance check yet

The MIMIC causes/indicators framework Layer D borrows its variable list from (Section 6).

- [8] Siddiqi, N. (2012). *Credit Risk Scorecards: Developing and Implementing Intelligent Credit Scoring*. Wiley. Weight-of-evidence scorecard construction (Section 4).

A Reproducibility Map

Every formula in this paper is a direct transcription of running code, not a description of intended behavior. Any figure may be verified by opening the file it names.

Section	File
3 — forensic signals	<code>src/lib/forensics.js</code>
4–5 — scorecard, confidence	<code>src/lib/shadow-score.js</code>
6 — regional macro context	<code>backend/lib/mimic-data.js</code>
7 — prompt and schema	<code>backend/lib/prompt.js</code> <code>backend/lib/gemini.js</code>
7 — usage logging	<code>backend/lib/usage-logger.js</code>

Shadow Index — Working Paper v3. Synthetic demonstration data throughout. Deterministic scorecard live; audit-label calibration not yet run.